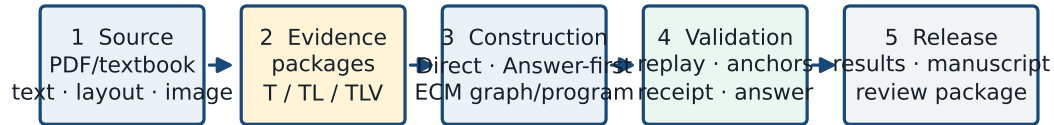


ECM-TQAG data pipeline



Frozen evidence → typed construction → deterministic replay → documented release